

BRAC University

Department of Computer Science and Engineering

CSE111: Programming Language II

Examination: Quiz #3

Date: 17 / 08 / 2022

Semester: Summer 2022

Time: 40 Minutes

ID: _____	Name: (Please write in CAPITAL LETTERS)	Obtained Points: / 20	Section: 42
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- Read questions carefully.
- Understanding the question is part of the exam, please do not ask questions.

1. The heroes of Hawkins fought very hard to fight against the monster Vecna. They were able to save their family and friends from being crumbled by Vecna. However, things got worse since Loki - son of Odin - killed He Who Remains (HWR) and invaded TVA. Multiple realities have been affected since the door of the multiverse has been opened. As a result, it took a while but Vecna was successful in escaping the Upside Down dimension and came to our dimension.

Vecna's target is no more limited to only the people of Hawkins but the whole globe. Besides being a Ph.D. holder in the English language, and being trapped for about 30 years alone, he has mastered Computation and Networking. To begin with, he challenged the programmers by broadcasting a piece of **driver code** along with an expected console-based **output**. The **broadcast** has proof that he was, is, and always will be the only one of himself so, there cannot be more than one Vecna. Now, it is the responsibility of programming enthusiasts - like you - to reverse engineer and find out if there are any hints of his next plan so that leaders of the world can come together and fight against this ruthless monster and save our reality. In fact, if the right steps are not taken, this could become a multiversal crisis because Vecna still is not aware of the mess Loki created. Therefore, you must hurry and solve the problem within the next **25 minutes** to minimize the losses.

Additionally, your teacher - knowing that you are doing such amazing work - looked into the matter and gave you a hint that Vecna might be targeting every object of a Person right at creation time but only crumbles those whose name has at least 3 vowels. In addition to that, your teacher has also provided you with the below class diagram which might help you understand the problem further. (10 marks)

Vecna
<u>- instances: Vecna</u> + targets: [] + crumbles: []
+ __init__(): None + <u>cast(): Vecna</u> + next(): Person + <u>greet(): None</u> + details(): None

Person
+ name: str
+ __init__(): None + __repr__(): str

Driver code:

```
# Write your code here

james = Person("James")
robert = Person("Robert")
john = Person("John")
david = Person("David")
james.name = "James Matthew"
vecna = Vecna.cast()

victim1 = vecna.next()
Vecna.details(vecna)
print("+++++")
print(john == vecna.next())
victim3 = vecna.next()
vecna.details()
```

Output:

```
Tik! Tok! James Tik! Tok!
Tik! Tok! Robert Tik! Tok!
Tik! Tok! John Tik! Tok!
Tik! Tok! David Tik! Tok!
Dear James Matthew, it's time to free your suffering.
=====
Tik Tok! Tik! Tok!
Targets: [Robert, John, David]
Crumbles: [James Matthew]
+++++
No one is suffering, Hmmm!
False
No one is suffering, Hmmm!
=====
Tik Tok! Tik! Tok!
Targets: [Robert, John, David]
Crumbles: [James Matthew]
```

2. Answer any of the two questions below, both for the bonus.

(10 Marks - each)

Note: Bonus will be given if at least one question is answered correctly. Otherwise, the highest number obtained will be considered and no bonus will be given.

- a. Given the Tariff Rate below - retrieved from: <https://dpdc.org.bd/page/view/18>, along with the expected output and a driver code of DPDC class. Your task is to reverse engineer and design the **DPDC** class in Python without modifying any parts of the driver code to generate the exact given output. Be noted that DPDC class has the following class variables: **count** and **instances**, and each object of the DPDC has the private attributes: **id** (auto-generated) and **area**; where both ID and area are readable but only area is writable for valid (str) values.

SL	Customer Category and Slab	New Tariff Per Unit Rate (Tk.) Effective from Bill month September, 2015
1	Category A: Residential	
	Life Line: 1-50 Units	3.33
	a First Step: From 00 to 75 units	3.80
	b Second Step: From 76 to 200 units	5.14
	c Third Step: From 201 to 300 units	5.36
	d Fourth Step: From 301 to 400 units	5.63
	e Fifth Step: From 401 to 600 units	8.70
	f Sixth Step: From 601 to above	9.98

Driver code	Expected Output
<pre># Design the DPDC class here dhanmondi = DPDC("Dhamondi") narayanganj = DPDC("Narayanganj") damra = DPDC("Damra") dhanmondi.id = 111 dhanmondi.area = "Dhanmondi" print(dhanmondi.id) DPDC.stats() units = 23, 41, 79, 192, 512 for unit in units: # for each unit in units: print(DPDC.residential_step_tariff(unit))</pre>	<pre>1 { 1: Dhanmondi } { 2: Narayanganj } { 3: Damra } 87.39999999999999 155.79999999999998 406.06 986.8799999999999 4454.4</pre>

- b. The last 25 minutes were tough. Now relax and trace the following Python program for **10 marks** with PeaceMachine. And before submitting for proof-check, make sure that you have shown the output along with your workings in the tracing table.

1	class PeaceMachine:
2	__total = 0
3	def __init__(self, name):
4	self.name = name
5	PeaceMachine.__total += 1
6	self.pid = self.__total
7	
8	def maintain_peace(self):
9	print("Pease shall remain...", end=" ")
10	return self
11	
12	@classmethod
13	def bring_peace(cls, name):
14	return cls(name)
15	
16	@classmethod
17	def peace_reborn(cls):
18	cls.__total = 0
19	return cls
20	
21	@staticmethod
22	def write(msg):
23	print(msg, "< Peace Writer >")
24	
25	cse111 = PeaceMachine.bring_peace("CSE111")
26	cse111.maintain_peace().write("No matter what!")
27	PeaceMachine.write("Powered by")
28	cse221 = PeaceMachine("CSE221")
29	print(cse221.name, cse221.pid)
30	print(cse221.peace_reborn().bring_peace("CSE220").pid)